NN-P-73A
December 2, 1969
SUPERSEDING
Int. Fed. Spec. NN-P-0073(GSA-FSS
July 11, 1966

FEDERAL SPECIFICATION

PALLETS, MATERIAL HANDLING, PLYWOOD, DOUBLE FACED STRINGER AND BLOCK CONSTRUCTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two-way entry and four-way entry double deck reversible and non-reversible pallets made from plywood or a combination of plywood and conventional lumber.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. Pallets covered by this specification shall be of the following types, styles, sizes, kinds, assembly methods, and classes (conventional lumber), as specified (see 6.2):

Types:
- Type I - Two-way entry (stringer design).
- Type II - Four-way entry with rectangular wood spacer blocks.
- Type III - Four-way entry entry with laminated plywood spacer blocks.
- Type IV - Four-way partial entry - stringer design.

Styles:
- Style 1 - Full plywood deck, top, and bottom, reversible.
- Style 2 - Full plywood deck, top and bottom, with openings for hand trucks, nonreversible.
- Style 3 - Plywood top deck with three bottom plywood or conventional lumber deckboards, nonreversible.

Sizes: Unless otherwise specified (see 6.2), sizes shall be of the following lengths and widths:
- Size 1 - 32 by 40 inches.
- Size 2 - 40 by 48 inches.
- Size 3 - 48 by 48 inches.
- Size 4 - 42 by 52 inches.

Kinds:
- Kind 1 - Flush.
- Kind 2 - Single wing (overhang)
- Kind 3 - Double wing (overhang).

Assembly methods:
- Assembly method (a) - Nailed.
- Assembly method (b) - Bolted.

Classes:
- Class 1 - Group I low density woods.
- Class 2 - Group II and III medium density woods.
- Class 3 - Group IV high density woods.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

FF-N-105 - Nails, Wire, Brads, and Staples.

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specification and Standards from established distribution points in their agencies).

Military Specification:

MIL-W-6110 - Wood, Determination of Moisture Content Of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-731 - Quality of Wood Members for Containers and Pallets.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply:

Commerical Standard:

CS 35-61 - Hardwood Plywood.

Product Standard:

PS 1-66 - Softwood Plywood, Construction and Industrial.

(Copies may be obtained from the American Plywood Association, 1119 A Street, Tacoma, Washington 98401.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the applicable end item shall be submitted or made ready for approval by the contracting officer or his authorized representative in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

3.2 Materials.

3.2.1 Plywood. All plywood shall conform to Commercial Standard CS 35-61, grades 3 or 4, type I or Product Standard PS 1-66 STANDARD grade, bonded with exterior glue.

3.2.2 Lumber. All lumber used for stringers, deckboards, and blocks shall conform to the wood groups and quality requirements of MIL-STD-731 for pallets designed for multiple handling with the following exceptions. Lumber used shall be entirely free of Lyctus (Powder Post) beetle infestations as evidenced by fresh worm holes covered with fresh wood dust. All pallets found to have such infestation at time of Government inspection shall be rejected.

3.2.2.1 Moisture content. There shall be no limits on the moisture content of stringers.

3.2.2.2 Splits and shakes. Splits and shakes shall not be permitted in pallet blocks.

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3.2.2.3 Wane. Wane shall not be permitted in stringers and blocks.

3.2.3 Fasteners. Fasteners used to assemble pallets shall be of the following types:

3.2.3.1 Nails. Nails shall be either helically threaded or annular grooved conforming to FF-N-105, type II, style 18, 2-1/2 inches long, 0.120 inch wire diameter or type II, style 10, 2-1/2 inches long, 0.120 wire diameter, respectively.

3.2.3.1.1 Alternate nail. A suitable alternate to the style 18 nail conforming to FF-N-105 shall be a slender hardened drive screw nail meeting the following requirements:

Length: 2-1/4 inches
Wire diameter: 0.110 inch + 0.002 inch.
Threaded body outside diameter: 0.138 inch ± 0.005 inch.
Head diameter: 1/4 inch, minimum
Point: Diamond or chisel 5/32 inch long and not wider than the wire diameter.
Thread: Spiral minimum 4 flutes, helical angle of thread at the pitch diameter 60° ± 5° with a plane perpendicular to the axis.
Hardness: Rockwell C42 minimum.
Ductility: Sufficient to bend 90° about their own diameter without fracture.

3.2.3.2 Bolts and nuts. If bolts are used to assemble the pallets, they shall be of the type specifically designed for use with plywood having a flat head, as shown in figure 7. Bolts shall meet the following requirements:

Length: 4-1/2 inches.
Diameter: 5/16 inch.
Thread: 5/16 inch - 18-NC-2.
Head diameter: 1 inch + 1/32 inch.
Head thickness: 0.031 inch to 0.063 inch at periphery, 1/8 inch at root.
Anti-rotational devices: There shall be a minimum of two protrusions on the underside of the bolthead, either formed or punched, to prevent the rotation of the bolt during the tightening of the nut. Nuts shall be of the "T" type (Allen or other suitable type) and at least 1 inch in diameter.

3.3 Details of components.

3.3.1 Plywood deck (including deckboards). Unless otherwise specified (see 6.2) plywood decks and plywood bottom deckboards shall be 5/8-inch thick. The plywood deck shall consist of a single piece of plywood.

3.3.2 Pallet stringers. Pallet stringers shall be surfaced top and bottom to a uniform height and be of nominal 2 x 4 inch dimensions.

3.3.3 Wood blocks. Rectangular wood blocks shall be surfaced top and bottom to a uniform height, and shall have nominal dimensions of 4 by 4 inches and shall be 6 inches long for size 1 and 2 pallets, and nominal 4 x 6 inches and shall be 6 inches long for size 3 and 4 pallets. Outer faces of all outside blocks shall be finished and ends of blocks shall be cut square with sides.

3.3.3.1 Cylindrical plywood spacer blocks (plyblocks). Plywood spacer blocks shall be a minimum of 3 inches in diameter. Blocks shall not vary from the specified height of 3-5/8 inches by more than 1/16 inch.

3.3.3.2 Rectangular plywood spacer blocks. When bolted pallets are furnished, rectangular laminated plywood spacer blocks may be supplied in lieu of the solid blocks required in 3.3.3 above, provided the blocks are built up from plywood laminated with exterior glue and meeting the proper grademark requirements of the appropriate commercial or product standard. The builtup blocks shall be constructed of exterior grade plywood, held together with 2-3/4 inch minimal height staples.

3.3.4 Bottom (lumber) deckboards. Bottom deckboards shall have the exposed surfaces finished smooth and shall extend the full width of the pallet. Deckboards shall be nominal 1 inch lumber of uniform thickness. Outside or end bottom deckboards and the center deckboard of block type pallets shall have a nominal 6 inch width; the center deckboard of stringer type pallets shall have a nominal 8 inch width.

3.3.4.1 Bottom (plywood) deckboards. If plywood is specified for bottom deckboards (see 6.2), width dimensions shall conform to those specified in 3.3.4 for conventional lumber deckboards.

3.3.5 Chamfering. The bottom deck of nonreversible pallets shall have 45 degrees 1/4 inch minimum length chamfers in locations specified on figures 2, 3, 4, and 6.

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3.3.6 Bull-nosing. Top and bottom plywood decks shall be either square edged or bull-nosed (beveled) as specified (see 6.2).

3.3.7 Rounded corners. Top and bottom plywood decks shall have corners rounded or squared as specified (see 6.2).

3.4 Fabrication. Pallets shall be fabricated as specified herein and as shown on figure 1, 2, 3, 4, 5, and 6, as applicable.

3.4.1 Assembly. Pallets shall be assembled using any of the mechanical fasteners herein specified unless only one fastener type is specified (see 6.2), except that slender hardened drive screw nails as described in 3.2.3.1.1 shall be used only for securing lumber bottom deckboards. Spacing of nails shall be shown on figures 1, 2, 3, 4, 5, and 6, as applicable.

3.4.2 Wood grain. The plywood face grain shall be placed at right angles to the stringers. In rectangular block pallets the grain shall be placed parallel to the longest span between blocks. All blocks shall have side grain (not end grain) in contact with the plywood decks. The face of the grain of bottom (plywood) deckboards shall be parallel with the length of the deckboard.

3.4.3 Construction. Pallets shall be accurately formed to require dimensions with all joints tight. No fasteners shall protrude above the surface of the wood.

3.4.3.1 Nailed pallets. Nails may be machine or hand drive. All nails bent in driving shall be removed or broken off below the surface and replaced. Nail locations shall be as indicated by the figures herein to minimize splitting. When pallets are nailed by hand, predrilling is permitted and when utilized shall be accomplished with a 3/32 (0.0937) inch diameter drill.

3.4.3.2 Bolted pallets. Two-way entry or partial four-way entry, three stringer pallets shall be assembled with a minimum of nine bolts with three bolts per stringer. A fully reversible pallet requires no additional fasteners; however, a nonreversible pallet shall have the bottom tie members, which do not have a bolt passing through them, fastened to the stringers with a minimum of two nails per stringer contact point. Four-way entry pallets of the block type shall have at least one bolt per block. In addition to the bolts, nails or other suitable anti-rotation devices shall be installed to prevent the rotation of the blocks during in-service use. If cylindrical blocks are used, an anti-rotation device is not necessary in addition to the bolt.

3.4.3.2.1 Bolt holes. Bolt holes shall be drilled 1/32-inch in diameter greater than the size of the bolts being used to assemble pallets. The drilling shall be such as to produce a smooth bolt hole. The depth of countersink for each bolt shall be such that the strength of the pallet will not be impaired.

3.5 Tolerances. A plus or minus tolerance of 1/4-inch will be permitted on overall width and length dimensions of pallets. Out of square deviation shall not exceed 1/4-inch difference in diagonal measurement. Where minimum dimensions are specified, no minus tolerance will be permitted.

3.6 Marking. When specified (see 6.2), pallets shall be branded in letters 1-inch to 1-1/2-inches high on the outside face of one of the outside stringers or blocks with the letters "U.S". The manufacturer's identification shall be neatly stenciled on the same stringer or block face in letters 1/4- to 1/2-inch high. Color coding, if required, shall be as specified (see 6.2).

3.7 Workmanship. Workmanship shall be in accordance with the best grade of commercial practice. Pallets shall be clean and properly assembled. Pallet surfaces shall be smooth and free from silvers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required (see 3.1), it shall be examined for defects in 4.3.3.1, dimensional requirements specified herein, and tested for moisture content in accordance with 4.4.1.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with

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requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.3.2 Intermediate testing. Intermediate testing shall consist of testing for moisture content in accordance with 4.4.1. The sample unit shall be one conventional lumber deckboard. Sample units shall be obtained from each lot of conventional lumber deckboards used in constructing the pallets. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

4.3.3 End item inspection. The lot size shall be all pallets of the same type, style, size, kind, assembly method, and class submitted for acceptance at one time. The sample unit shall be one assembled pallet.

4.3.3.1 Visual examination. Examination of the end item shall be in accordance with the classification of defects listed in table II. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects.

Examine	Defects	Classification Major	Classification Minor
Conventional lumber	Not class (species) specified.	X	
	Not free of lyctus powder post beetles infestation.	X	
	Not free of decay.	X	
Stringers and conventional lumber blocks	Knots, knotholes, loose knots, and knot clusters, and holes having a diameter greater than 3/8 the width of the wood member.	X	
	Through shakes and splits longer than the width of the wood members.	X	
	Pith not firm and tight.	X	
	Presence of shakes and splits in blocks.	X	
	Grain slope greater than 1 in 10.	X	
	Presence of wane.	X	
Bottom deckboards (conventional lumber)	Knots, knotholes, loose knots and knot clusters and holes other than knots having a diameter greater than 3/8 the width of the wood member.	X	
	Pith not firm and tight.	X	
	Wane greater than the following:		
	Bow - 1/8-inch per foot of length.	X	
	Crook - 1/16-inch per foot of length.	X	
	Cup - 1/4-inch per 8-inch width or like proportion.	X	
	Twist - 1/4-inch per 8-inch widths or like proportion.	X	
	Wane exceeding 3/4 the thickness or 1/4 the width of the member	X	
	Chipped or torn grain over 1/16-inch deep and exceeding an aggregate area 1/6 the surface area of the piece.	X	
	Slope of grain steeper than 1 in 10.	X	
	Not surfaced where required.		X
	Not chamfered as required.	X	
Plywood	Not grade specified.	X	
	Not fabricated with exterior glue.	X	
	Laminations separated.	X	
Fasteners	Nails or bolts not the type specified.	X	
Assembly	Assembled pallet does not form a rectangle with 90 degree corners.	X	
	Stringer protrudes on any side.		X
	Bolted pallets with rectangular blocks do not have anti-rotation devices applied to the blocks.		X

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Assembly (con)	Fasteners protrude above the surface of the wood.	X	
	Nails not staggered or positioned as specified.	X	
	Components assembled with wood grain not in specified direction.	X	
	Joints not tight.		X
Marking (and branding when applicable)	Type, size, and location not as specified; not legible and not permanent.		X
	Not color coded when required.		X

4.3.3.2 Dimensional examination. Pallets shall be examined for compliance with specified dimensions. Any noncompliance with specified dimensions shall constitute a defect. The inspection level shall be S-2 with an AQL of 6.5 expressed in terms of defects per hundred units.

4.4 Tests.

4.4.1 Moisture content of wood. Conventional lumber deckboards shall be tested for moisture content to determine conformance to 3.2.3. Determination of moisture content shall be made using the Electric Moisture Meter Method of MIL-W-6110. In case of dispute the moisture content shall be verified by the Oven Drying Method of the above specification.

5. PREPARATION FOR DELIVERY

5.1 Shipment. Unless otherwise specified in the contract or order (see 6.2), shipment shall be made in carload or truckload lots where quantity permits. Shipments shall be in accordance with best commercial practice, and offered in such manner as to insure acceptance by common or other carrier for safe transporation at the lowest rate to point of delivery. The loading of assembled pallets in freight cars and trucks shall be in such a manner as to facilitate unloading by use of forklift equipment. Nonreversible pallets shall be loaded with top deck in "UP" position.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order (see 6.2), shipments shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order (see 6.2) shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Pallets covered by this specification are intended for use with powered forklift trucks and manually operated pallet lift truck equipment for general materials handling operations in storage and distribution systems. They are used to palletize such commodities as bags, bales, barrels, boxes, cartons, cases, drums, kegs, sheeting, tubes, and wire, for world wide shipment and storage.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, size, kind, assembly method, and class required (see 1.2).
(c) Size required, if different from 1.2.
(d) When a preproduction sample is required (see 3.1).
(e) Thickness of plywood decks and plywood bottom deckboards if different from 3.3.1.
(f) When bottom deckboards are to be made of plywood (see 3.3.4.1).
(g) Whether top and bottom plywood decks shall be square edged or bull-nosed (bebeled) (see 3.3.6).
(h) Whether top and bottom plywood decks shall have corners rounded or squared (see 3.3.7).
(i) When pallets are to be assembled with only one type fastener and the type fastener to be used (see 3.4.1).
(j) When pallets shall be branded with the letters "U.S" (see 3.6).
(k) If pallets shall be color coded and the color coding required (see 3.6).
(l) Shipment lots if different from 5.1.
(m) Special marking when required (see 5.2.1 and 5.2.2).

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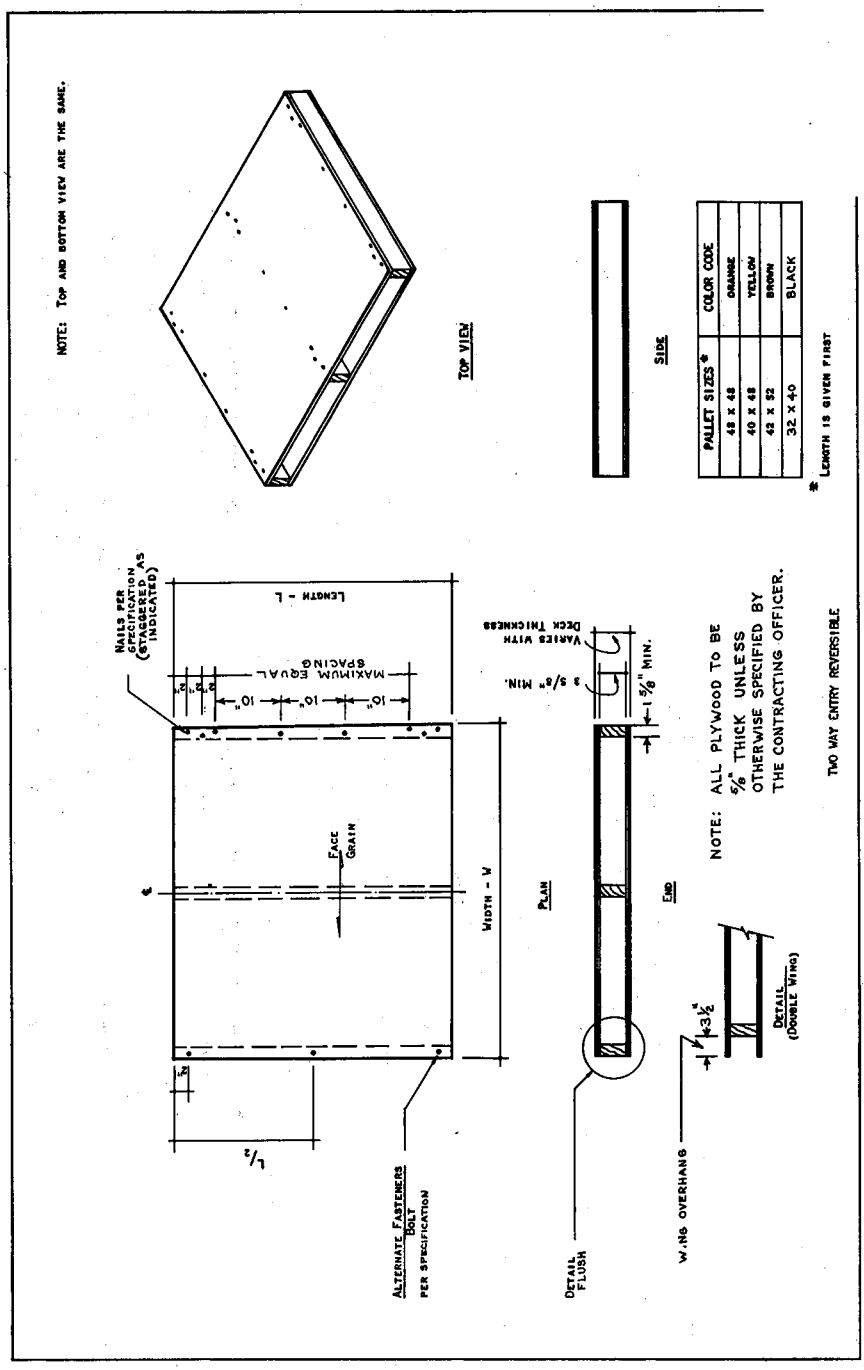


FIGURE 1

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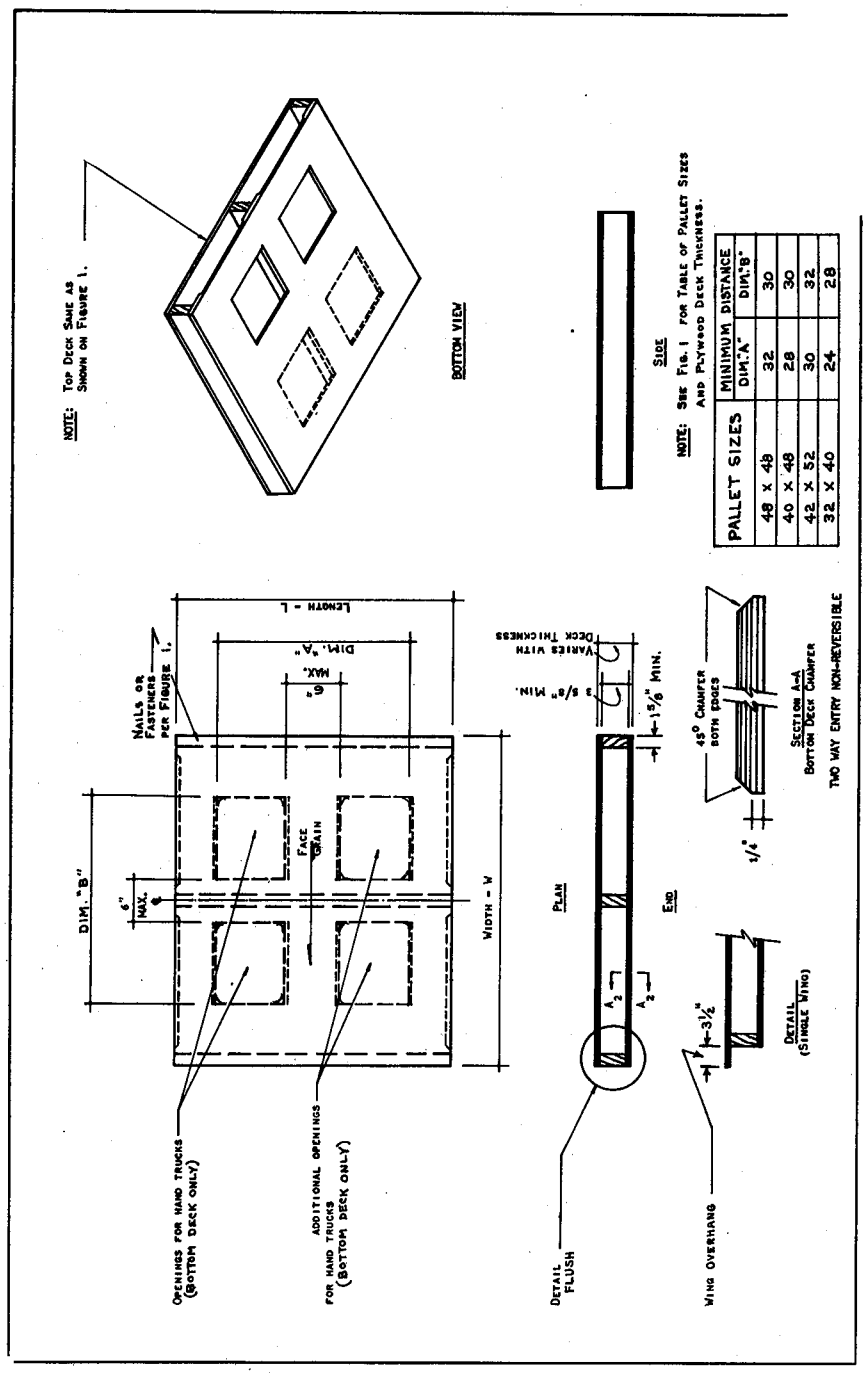


FIGURE 2

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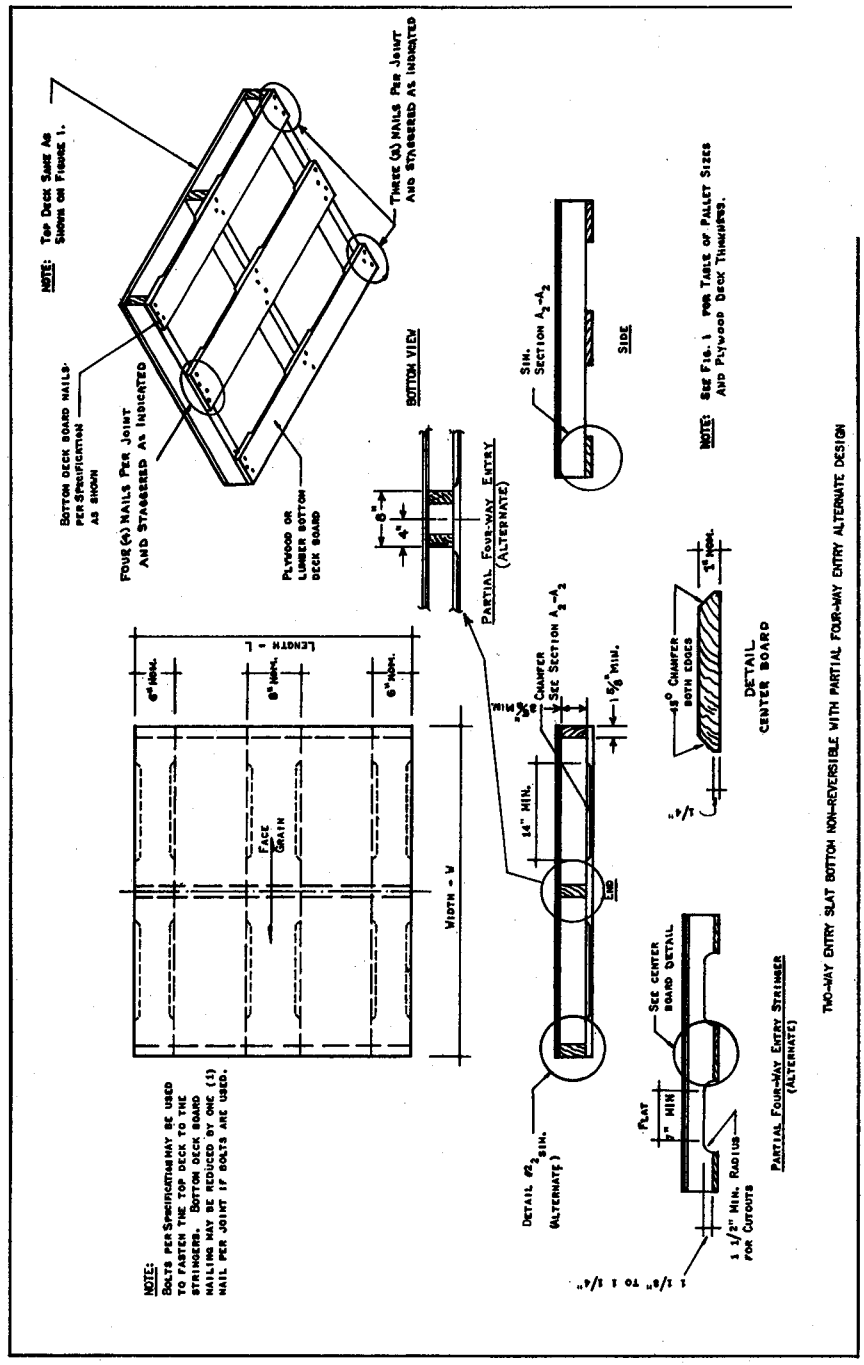


FIGURE 3

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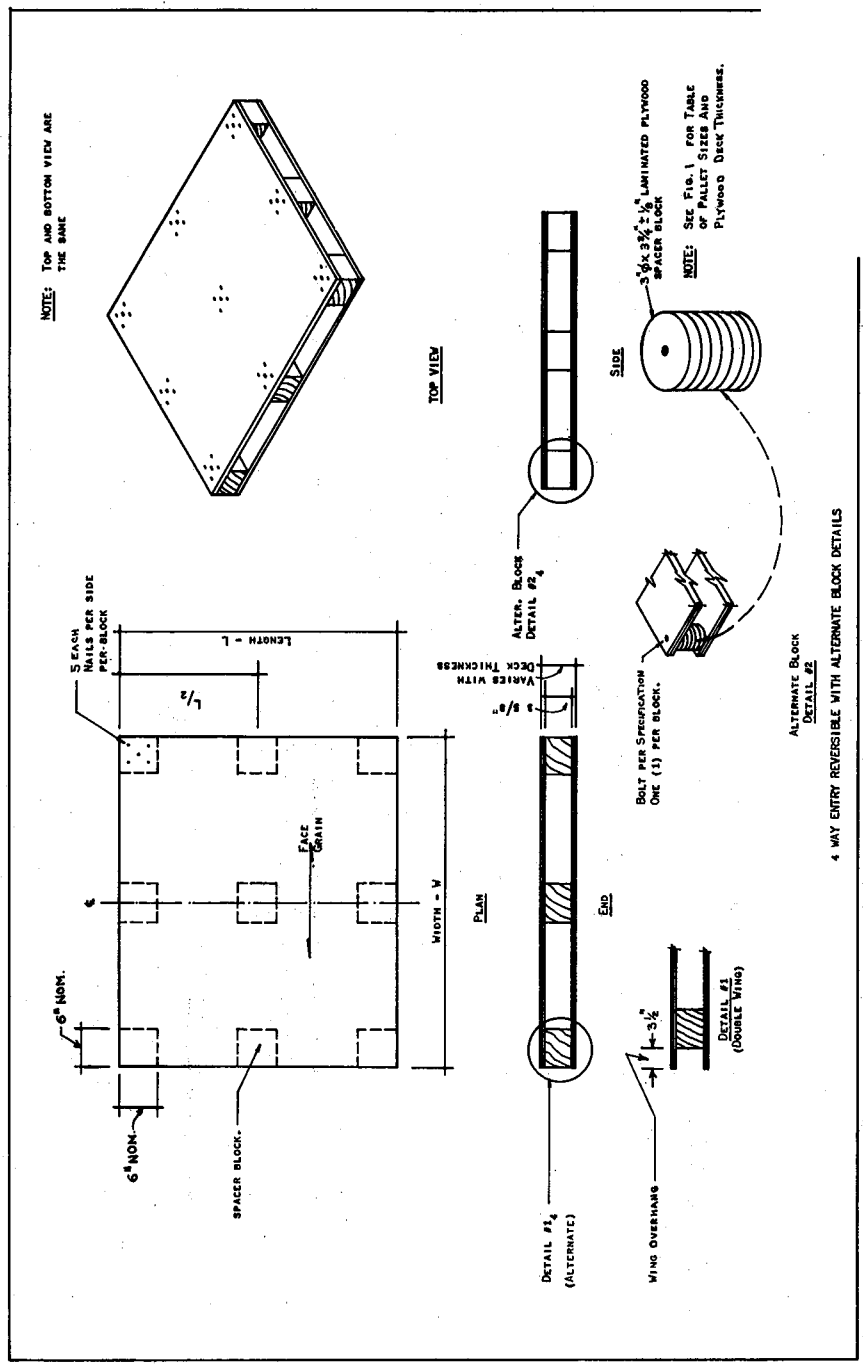


FIGURE 4

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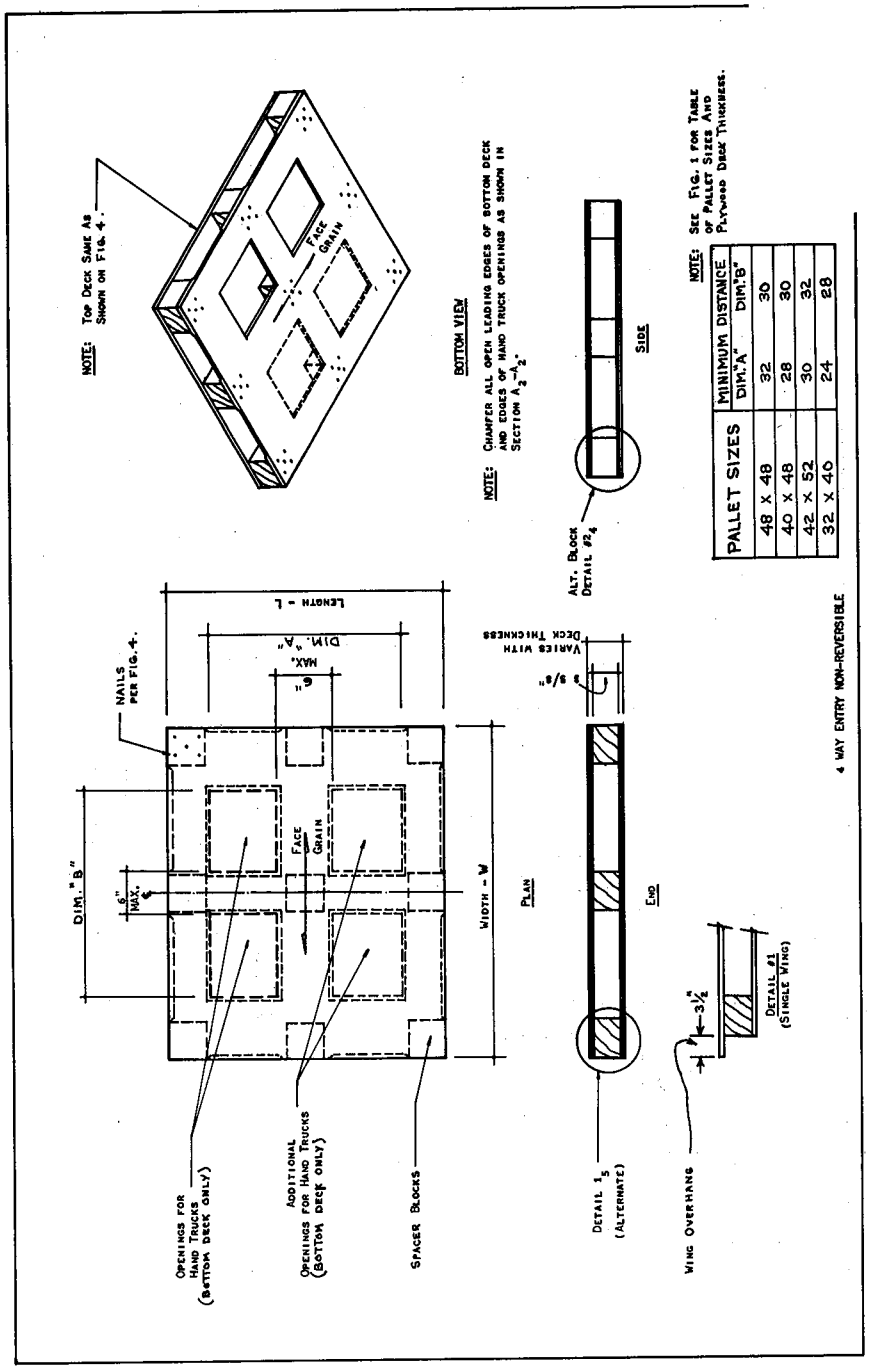


FIGURE 5

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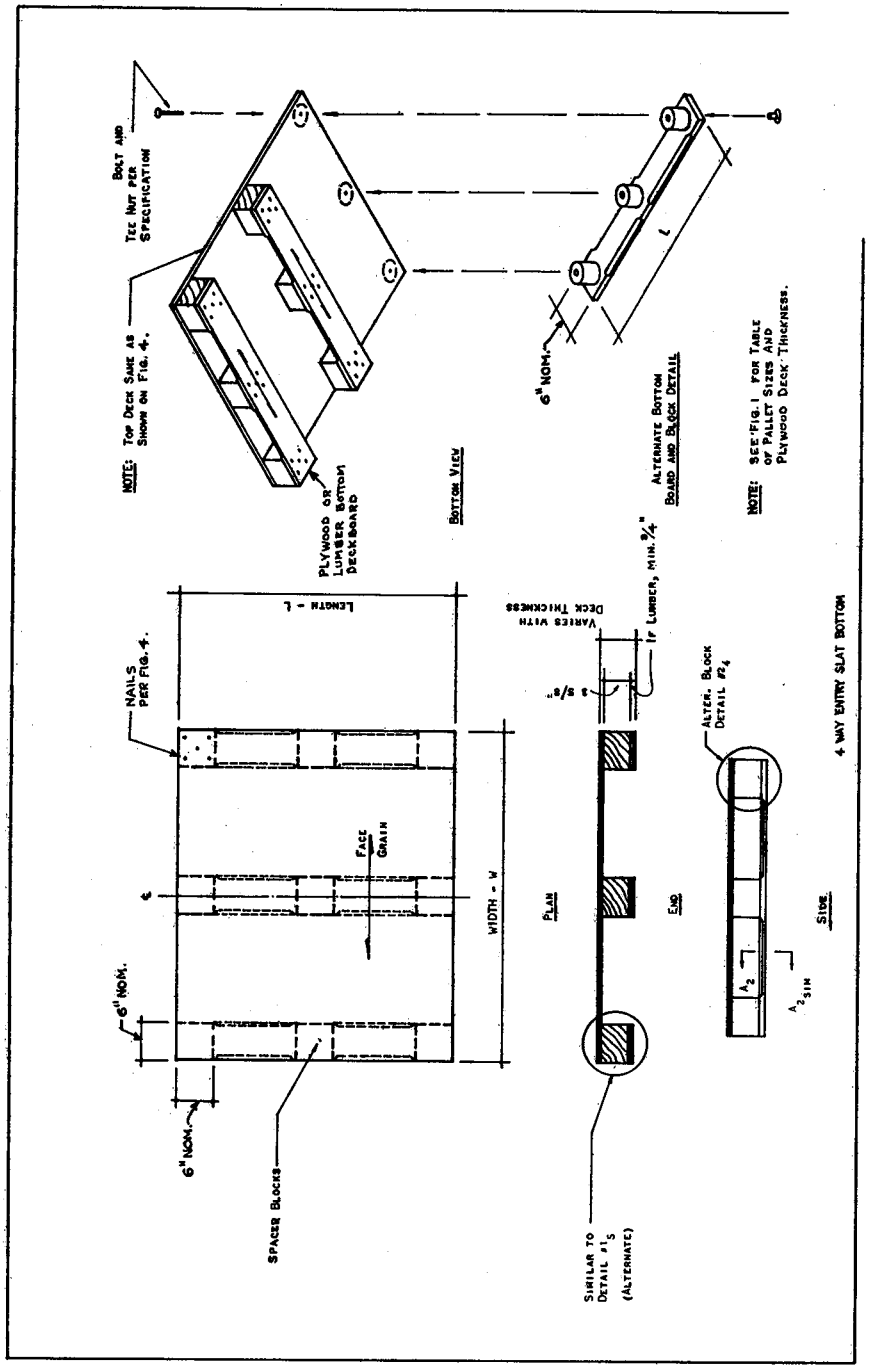


FIGURE 6

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

NN-P-73A Pallets, Material Handling, Plywood, Double Faced Stringer and Block Construction

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA-DC 69-9761

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GENERAL SERVICES ADMINISTRATION

General Services Administration
Federal Supply Service (FMSX)
Washington, D. C. 20406

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GSA FEB 69 2200

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